

```
%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;

% retrieve selected cuts used for further fitting
[GH_FFCutsFits,CutFitFileGH] =
recover_selected_cuts('GH_FFCutsFits','e:\SHARE\Fe\Data\analysis\07_fit_with_multicu
t_all\sel_cuts');
```

```
fn = fieldnames(GH_FFCutsFits);
%nf = 11
%for nf = 2:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei200_off200GHmaxFF';
if ismember(selectCuts(1:3),{'msk','dat'})||...
    contains(selectCuts,'all_fit_par')
    %continue
    error('Can not fit field: %s',selectCuts);
end

fprintf('*****\n');
```

```
*****
```

```
fprintf('Processing cuts: %s \n', selectCuts);
```

```
Processing cuts: Ei200_off200GHmaxFF
```

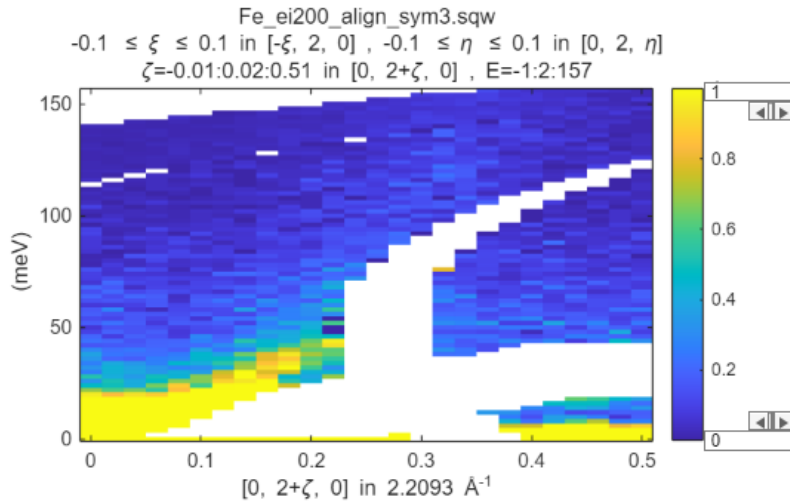
```
fprintf('*****\n');
```

```
*****
```

```
cuts_list = GH_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts,'Ei200')
    min_Ei = 30;
    mi = maps_instrument(200,600,'S');
end
if contains(selectCuts,'Ei400')
    min_Ei = 40;
    mi = maps_instrument(401,600,'S');
end
if contains(selectCuts,'Ei800')
    min_Ei = 70;
    mi = maps_instrument(786,600,'S');
end
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for j=1:numel(cuts_list)
```

```
cuts_list{j} = cuts_list{j}.set_instrument(mi);
cuts_list{j} = cuts_list{j}.set_sample(sample);
plot(cuts_list{j}); lz 0 1; keep_figure;
```

end



```
dir_name = "GH";

dE_step = 2; %original energy transfer step data were binned to. No point in going
finer
half_dE = 5; % half width of data binning

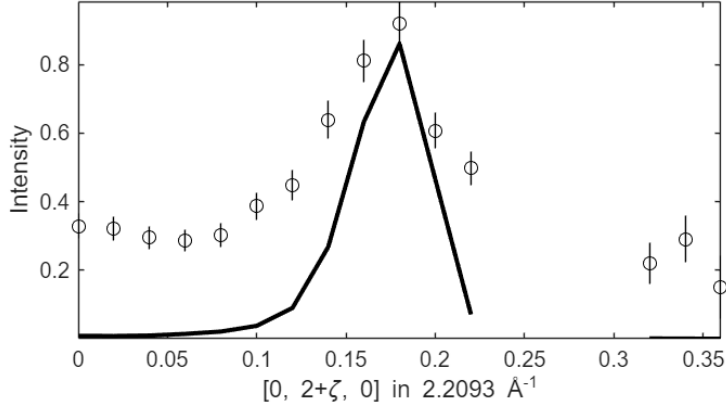
en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(min_Ei,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
GH_FFCutsFits.(selectCuts) = cuts_list;

GH_FFCutsFits = fit_multicutsFF_along_direction(...
    GH_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

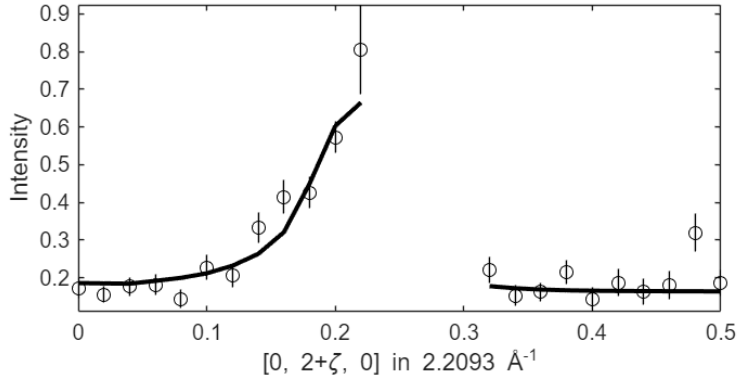
```
*****
**** En = 35±5 going to 145
*****
```

Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $30 \leq E \leq 4$
 $\zeta = -0.01:0.02:0.51$ in $[0, 2+\zeta, 0]$

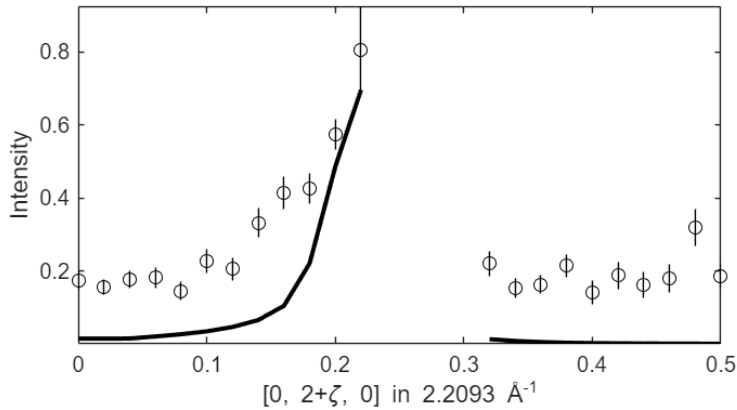


 **** En = 45±5 going to 145

Fe_ei200_align_sym3.sqw
 $S = 1.1 \pm 0.15$; $J_0 = 23 \pm 1.6$; $J_1 = 0 \pm 0$; $\gamma = 6.30636 \pm 0.538108$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $40 \leq E \leq 5$
 $\zeta = -0.01:0.02:0.51$ in $[0, 2+\zeta, 0]$



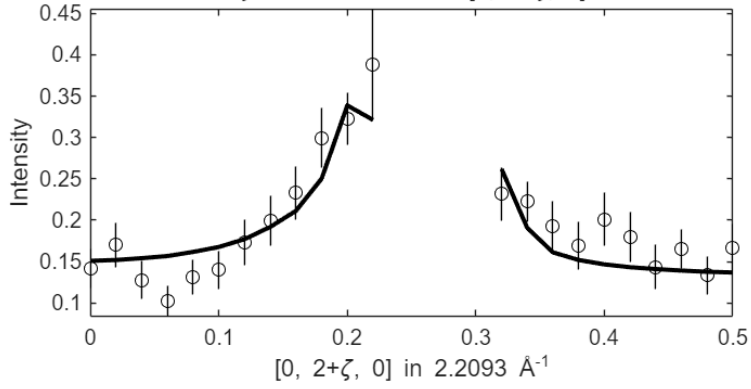
Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $40 \leq E \leq 5$
 $\zeta = -0.01:0.02:0.51$ in $[0, 2+\zeta, 0]$



 **** En = 55±5 going to 145

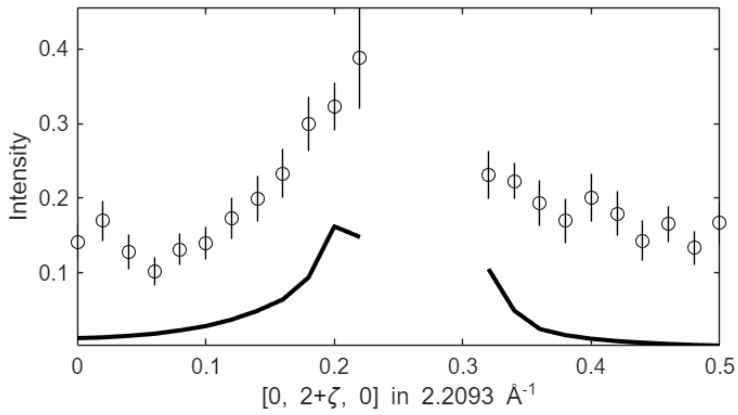
Fe_ei200_align_sym3.sqw

S= 1.6±0.34; J0= 21± 0.8; J1= 0± 0; gamma=9.40972±1.69433
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $50 \leq E \leq 60$
 $\zeta=-0.01:0.02:0.51$ in $[0, 2+\zeta, 0]$



Fe_ei200_align_sym3.sqw

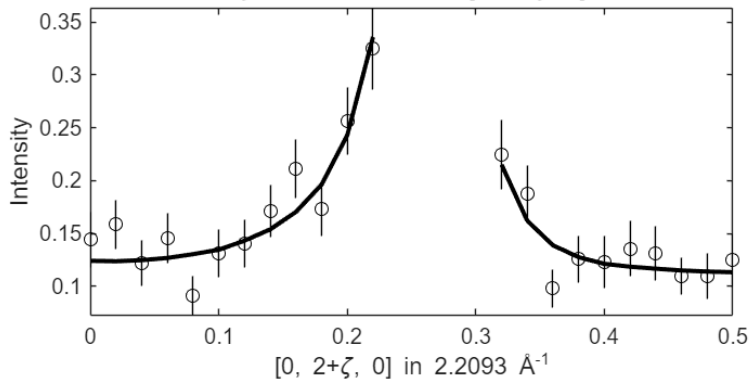
$-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $50 \leq E \leq 60$
 $\zeta=-0.01:0.02:0.51$ in $[0, 2+\zeta, 0]$



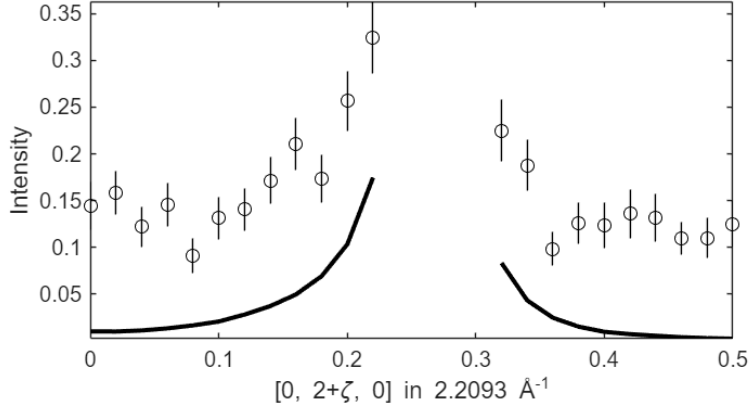
**** En = 65±5 going to 145

Fe_ei200_align_sym3.sqw

S= 1.6±0.43; J0= 24±0.74; J1= 0± 0; gamma=10.1356±2.58395
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $60 \leq E \leq 70$
 $\zeta=-0.01:0.02:0.51$ in $[0, 2+\zeta, 0]$

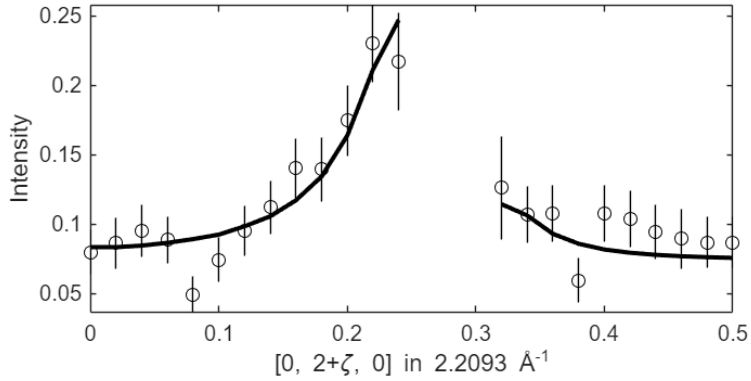


Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $60 \leq E \leq 7$
 $\zeta = -0.01:0.02:0.51$ in $[0, 2+\zeta, 0]$

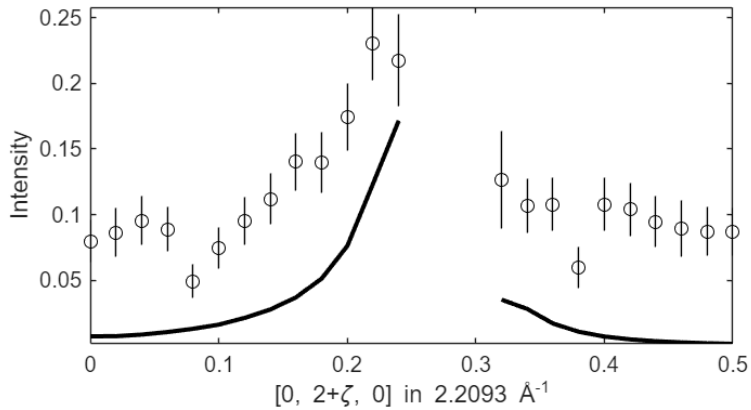


 **** En = 75±5 going to 145

Fe_ei200_align_sym3.sqw
 $S = 1 \pm 0.23$; $J_0 = 28 \pm 2.1$; $J_1 = 0 \pm 0$; $\gamma = 16.7604 \pm 2.14526$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $70 \leq E \leq 8$
 $\zeta = -0.01:0.02:0.51$ in $[0, 2+\zeta, 0]$



Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $70 \leq E \leq 8$
 $\zeta = -0.01:0.02:0.51$ in $[0, 2+\zeta, 0]$

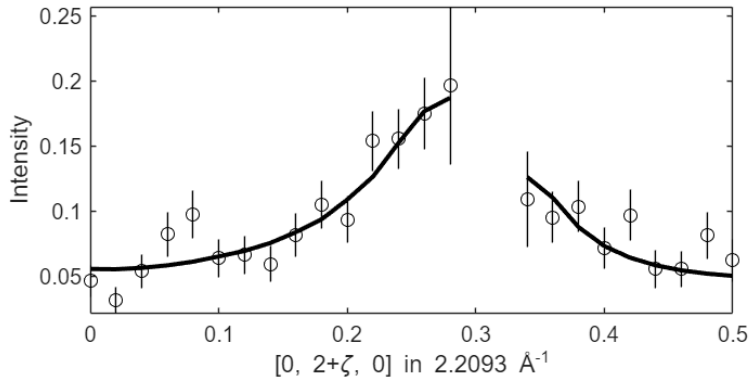


 **** En = 85±5 going to 145

Fe_ei200_align_sym3.sqw

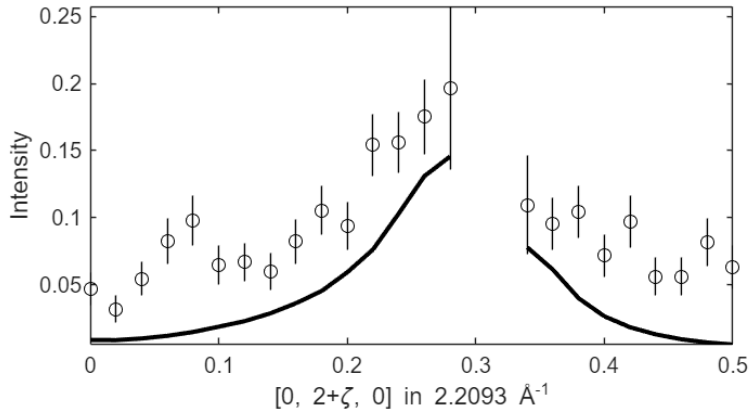
S= 1.3±0.28; J0= 27± 1.7; J1= 0± 0; gamma=30.4337±13.5128

-0.1 ≤ ξ ≤ 0.1 in [-ξ, 2, 0] , -0.1 ≤ η ≤ 0.1 in [0, 2, η] , 80 ≤ E ≤ 90
ζ=-0.01:0.02:0.51 in [0, 2+ζ, 0]



Fe_ei200_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [-ξ, 2, 0] , -0.1 ≤ η ≤ 0.1 in [0, 2, η] , 80 ≤ E ≤ 90
ζ=-0.01:0.02:0.51 in [0, 2+ζ, 0]

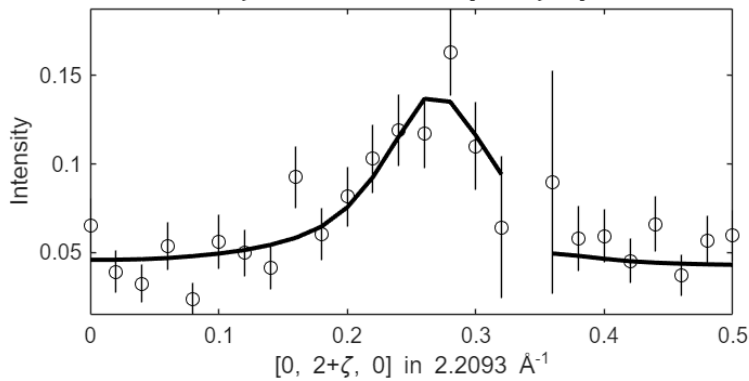


**** En = 95±5 going to 145

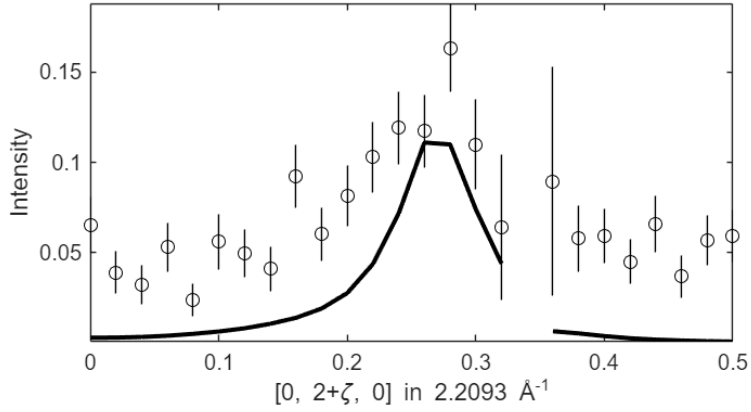
Fe_ei200_align_sym3.sqw

S=0.59±0.19; J0= 33± 1.9; J1= 0± 0; gamma=19.4133±10.2097

-0.1 ≤ ξ ≤ 0.1 in [-ξ, 2, 0] , -0.1 ≤ η ≤ 0.1 in [0, 2, η] , 90 ≤ E ≤ 100
ζ=-0.01:0.02:0.51 in [0, 2+ζ, 0]



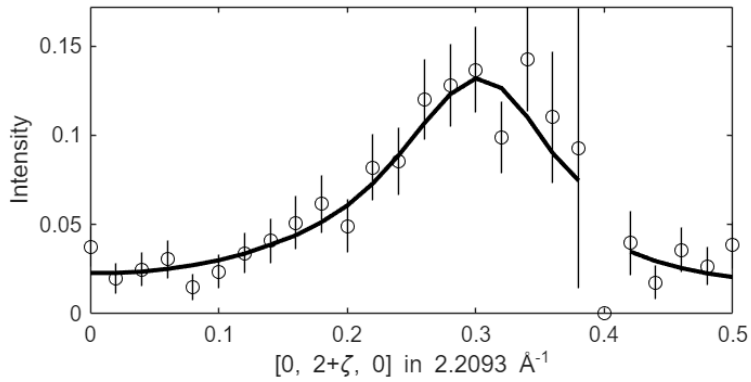
Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $90 \leq E \leq 110$
 $\zeta = -0.01:0.02:0.51$ in $[0, 2+\zeta, 0]$



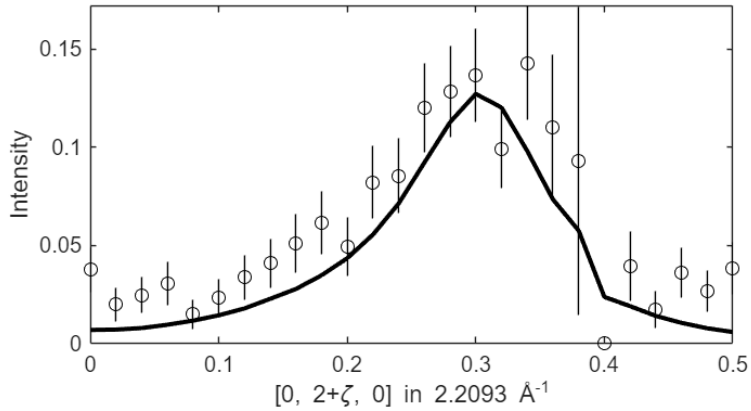
 **** En = 105±5 going to 145

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 3.8462

Fe_ei200_align_sym3.sqw
 $S = 1.4 \pm 0.24$; $J_0 = 31 \pm 1.3$; $J_1 = 0 \pm 0$; $\gamma = 41.8727 \pm 7.17828$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $100 \leq E \leq 110$
 $\zeta = -0.01:0.02:0.51$ in $[0, 2+\zeta, 0]$

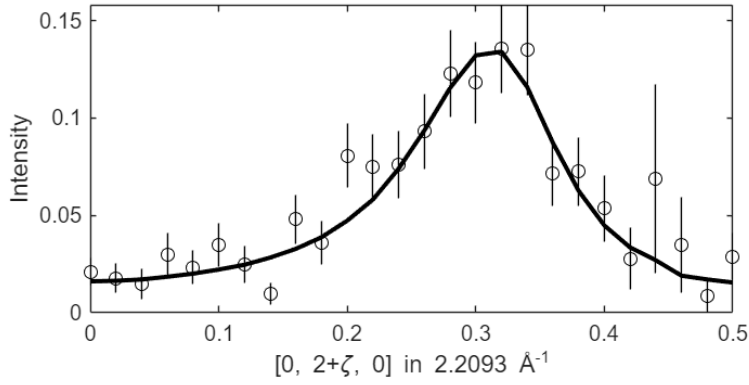


Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $100 \leq E \leq 110$
 $\zeta = -0.01:0.02:0.51$ in $[0, 2+\zeta, 0]$

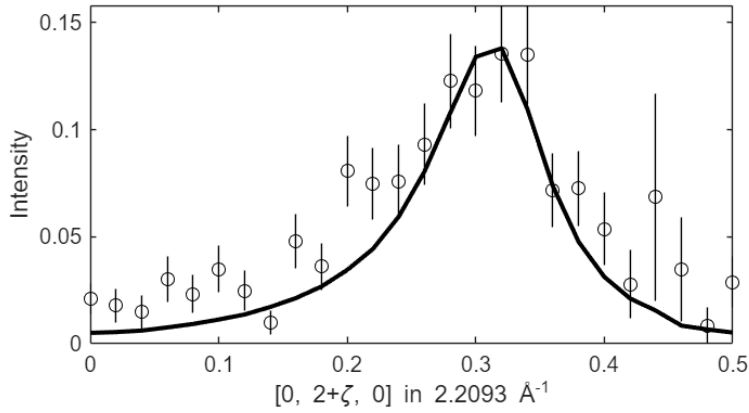


 **** En = 115±5 going to 145

Fe_ei200_align_sym3.sqw
 $S = 1.3 \pm 0.25$; $J_0 = 32 \pm 1.4$; $J_1 = 0 \pm 0$; $\gamma = 35.6805 \pm 9.24672$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $110 \leq E \leq 1$
 $\zeta = -0.01:0.02:0.51$ in $[0, 2+\zeta, 0]$

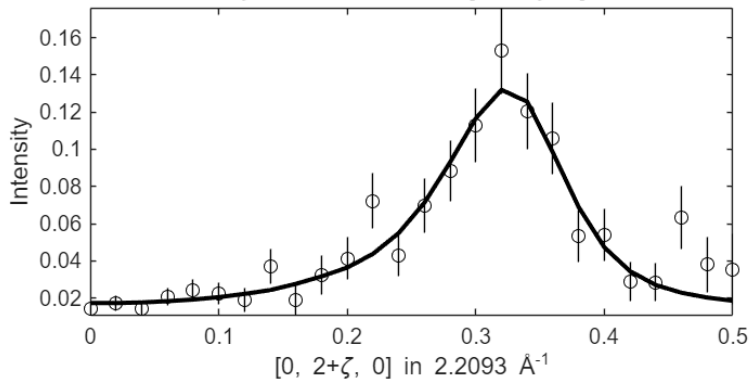


Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $110 \leq E \leq 1$
 $\zeta = -0.01:0.02:0.51$ in $[0, 2+\zeta, 0]$

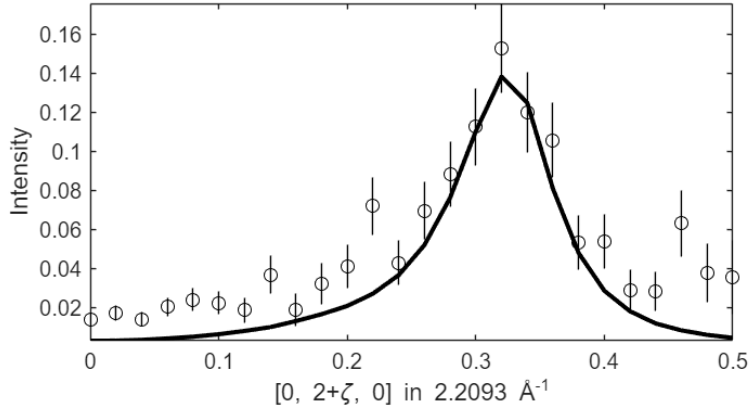


 **** En = 125±5 going to 145

Fe_ei200_align_sym3.sqw
 $S = 1.1 \pm 0.14$; $J_0 = 32 \pm 0.82$; $J_1 = 0 \pm 0$; $\gamma = 29.4043 \pm 6.1444$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $120 \leq E \leq 1$
 $\zeta = -0.01:0.02:0.51$ in $[0, 2+\zeta, 0]$

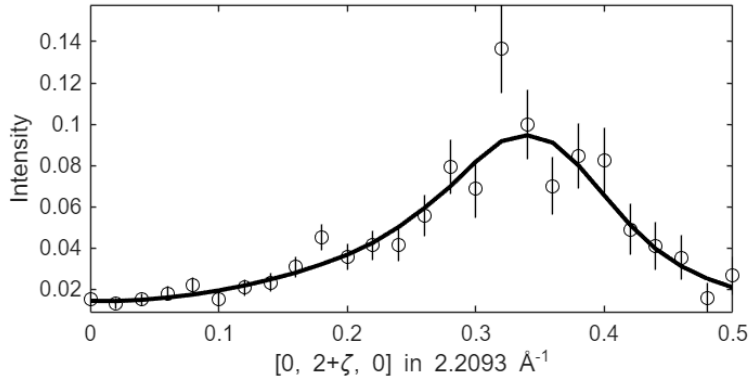


Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $120 \leq E \leq 1$
 $\zeta = -0.01:0.02:0.51$ in $[0, 2+\zeta, 0]$

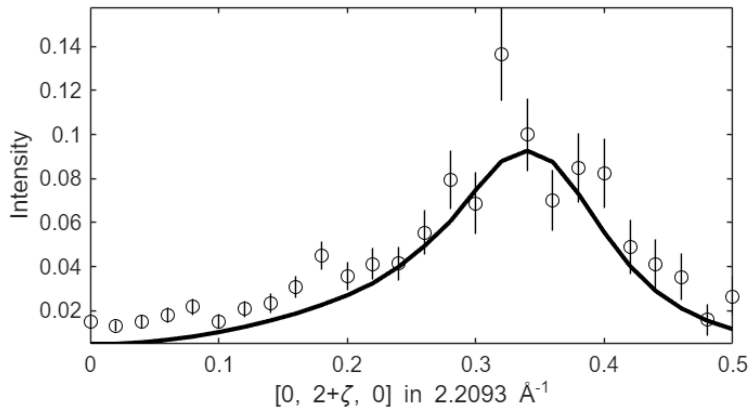


 **** En = 135±5 going to 145

Fe_ei200_align_sym3.sqw
 $S = 1.5 \pm 0.19$; $J_0 = 33 \pm 1.2$; $J_1 = 0 \pm 0$; $\gamma = 60.2685 \pm 11.6073$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $130 \leq E \leq 1$
 $\zeta = -0.01:0.02:0.51$ in $[0, 2+\zeta, 0]$

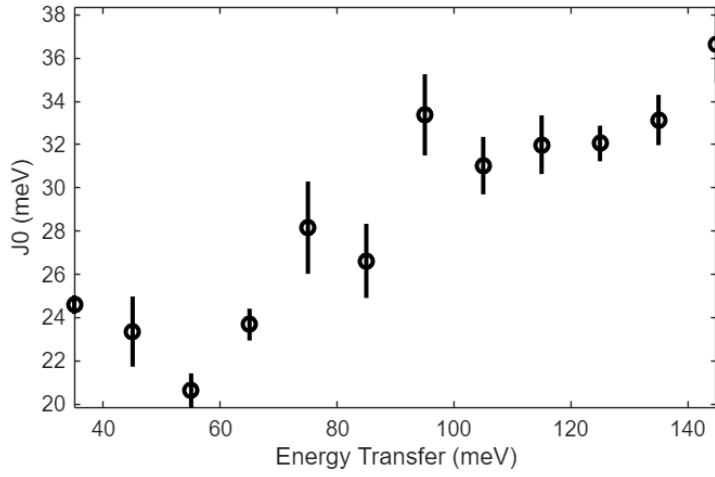
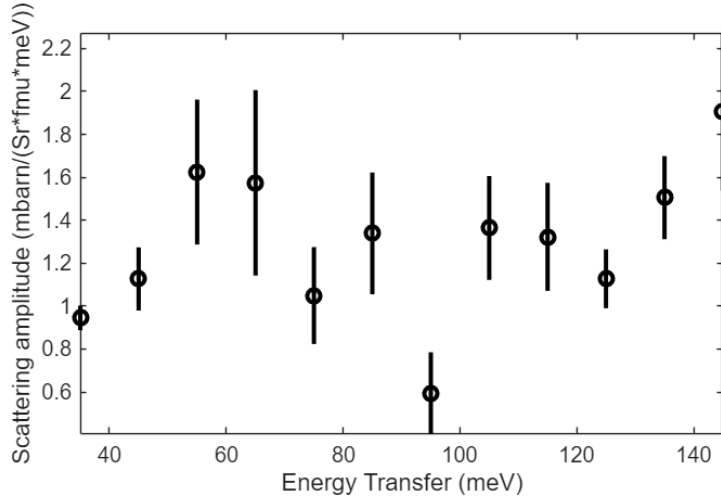
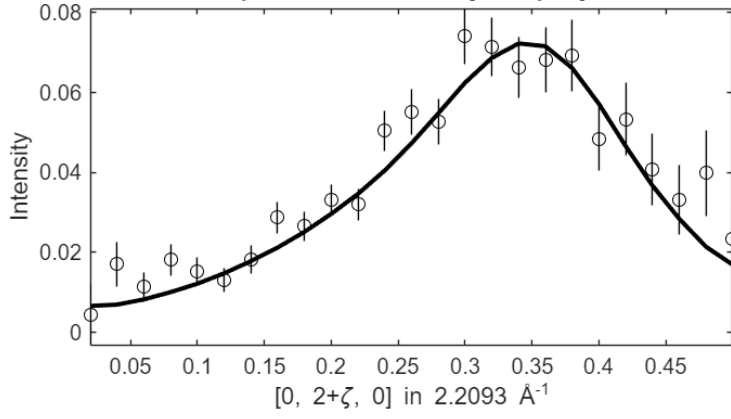


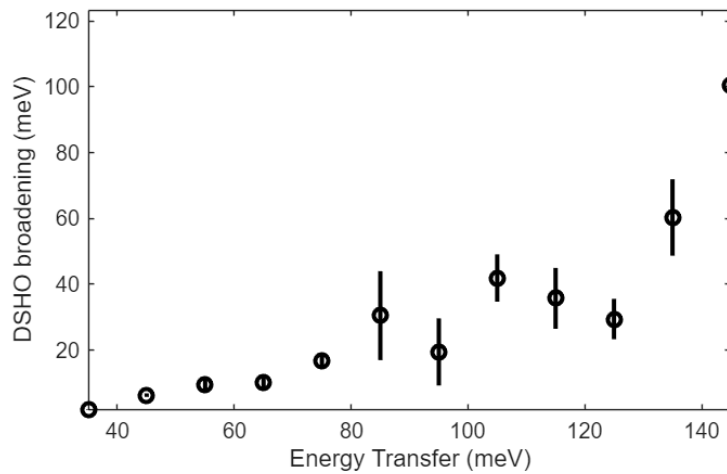
Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $130 \leq E \leq 1$
 $\zeta = -0.01:0.02:0.51$ in $[0, 2+\zeta, 0]$



 **** En = 145±5 going to 145

Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $140 \leq E \leq 1$
 $\zeta = -0.01:0.02:0.51$ in $[0, 2+\zeta, 0]$





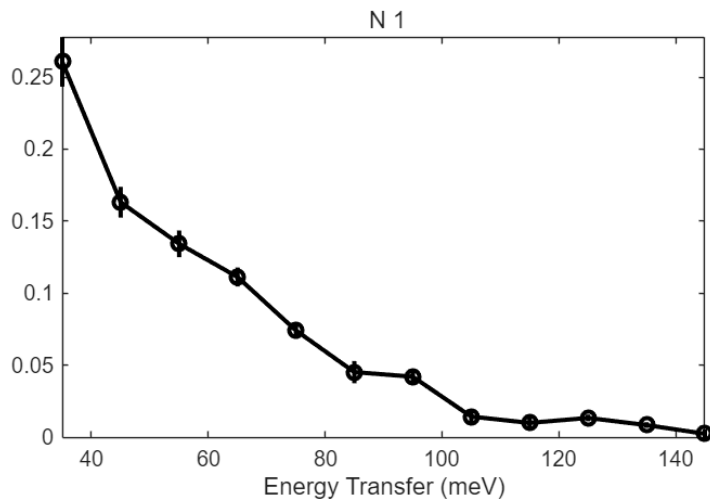
```

c1 = GH_FFCutsFits.(selectCuts);
GH_FFCutsFits.(selectCuts) = [c1{:}];

beep
beep
beep

all_fit_par = GH_FFCutsFits.([selectCuts, '_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;

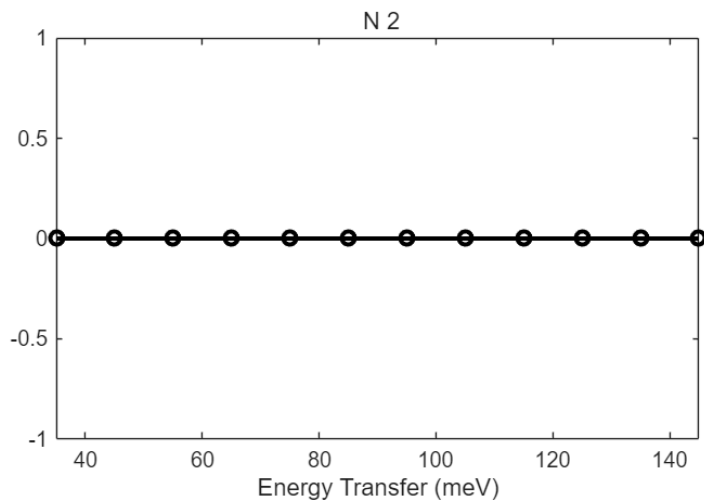
```



```

pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure

```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fig
ure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fig
ure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_GH.m';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_GH_',selectCuts,'.pdf']);
export(thisFile,targ_file);
```

```
fit_src_struct = GH_FFCutsFits;
save(CutFitFileGH,"fit_src_struct",-v7.3');
close all;
%end
```